HAB 3232 - Troubleshooting Guide

Revision H - 05/2026

Introduction

The following chapter is a troubleshooting guide for the Haas Automatic Band Saw 3232.

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- Symptoms Table
- Alarms Table
- Switch Troubleshooting
- Problem Solutions

⚠ **Only an authorized Haas Factory Outlet distributor should service or repair a Haas machine that is protected by the original factory warranty. Servicing by any other party voids the warranty.**

⚠ **Download and fill out the HAB3232 Troubleshooting Checklist below before replacing any parts.**

📄 **Note: If you cannot find the scenario that matches your situation, please contact service@haascnc.com.**

HAB3232 TROUBLESHOOTING CHECKLIST

Symptom Table

SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
The band saw blade is experiencing significant vibration.	Improper selection of saw teeth per inch.	Use the appropriate saw with the correct teeth per inch. Refer to section **4.5 HAB Saw Blade Selection** of the Operation chapter for more information.
	Broken teeth on the saw blade	Inspect blade and replace if necessary.
The saw blade is shifting during operation.	Saw blade tension.	Refer to the section **3.2 Blade Installation** of the Installation chapter for adjusting the tension of the blade.
	Carbides not secure on blade.	Black Knobs located on the blade arms must be loosely tightened to sandwich the blade between the carbide guides.
The saw blade is experiencing a reduction in cutting efficiency.	Lack of coolant.	Inspect the internal coolant tank and coolant pump. Clean metal chips from the coolant tank if needed.
Blade will not turn on.	No RPM set for operation.	When creating a material an rpm must be set or the blade will not turn on. The VFD will display a set frequency upon selecting a material.
The hydraulic pump will not start/stopped.	Low oil.	Fill with DTE 32 / 46 oil until it is more than 3/4 full.
	Hydraulic pump time out.	Hydraulic pump will turn off automatically after a set time.
Machine has no hydraulic pressure.	Machine is phased incorrectly.	Switch two legs of the input power to correct the direction of the the three phase hydraulic pump motor and the coolant pump motor.
The sight glass for either the coolant or the hydraulic oil appears to be leaking.	Sight glass has cracked.	If the sight glass is cracked, replace it with PN: **93-5207**.
Variable Frequency Drive intermittently shuts down.	VFD has been incorrectly wired.	Refer to section **6.5 HAB - Variable Frequency Drive Wiring** of the Troubleshooting chapter for the correction action
HAB does not permit the user to enter a wall thickness less than 10 inches.	Outdated Software.	Contact the Haas Service department for updated software. Visit the HAB Software Update for the proper installation instructions.
The band saw will not respond to any button commands.	Inspect the state of the Hybrid Amplifier.	Refer to the **DC Power Supply and Transformer** section below for the corrective action.
	System not initialized (V2 Saw only)	Return to main page and press button **'Saw Frame Return'** to initialize.
The Front Vise will only open incrementally when the frame is raised to its top travel position.	The proximity sensor has shifted out of position.	Refer to the **Proximity Height Sensor Section** below for the corrective action.
	The Proximity height sensor has failed.	
	Part length **had a uniform deviation from nominal.**	Kerf compensation not set properly. See section **4.8 HAB Cut Compensations** in the Operation Manual.
		Material Buffer parameter incorrect. See **Solenoid YV10 - Feed Buffer Parameter** section below.

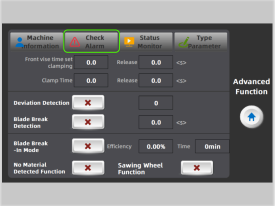
Part length is not accurate.	Part length is **varied and inconsistent.**	Mechanical Play. Check that the locknuts on the shuttle piston are secured.
		Blade tension too low. Increase the tension on the blade if necessary.
		Carbides not secure on blade. Black Knobs located on the blade arms must be loosely tightened to sandwich the blade between the carbide guides.
		Horizontal scale-encoder read issue. Machine is losing position at the ends of travel on the scale. Adjust the gap between the encoder and the scale to 0.030in.
The machine has started up in the wrong language.	Incorrect language setting.	Refer to the **How-To Change Language Settings** section below for the corrective action.
The saw blade cannot move downward despite being able to move up.	Material detection arm/sensor.	Loosen the knob below sensor SQ1 to allow the material detection arm to drop down. Ensure the red light for the SQ1 sensor turns on.

Alarms

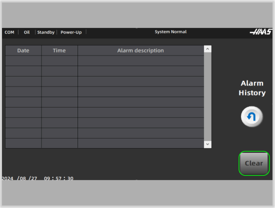
To check the list of alarms generated by the machine visit the Check Alarms page in the Advanced Functions tab.

The page will state the name and when the alarm was generated.

To wipe the alarm history press the Clear button.


Machine Information
Check Alarm
Status Monitor
Type Parameter
Advanced Function

Front vise time set clamping 0.0 Release 0.0 <S>
Clamp Time 0.0 Release 0.0 <S>
Deviation Detection ✗ 0
Blade Break Detection ✗ 0.0 <S>
Blade Break -In Mode ✗ Efficiency 0.00% Time 0min
No Material Detected Function ✗ Sawing Wheel Function ✗


COM
Oil
Standby
Power-Up
System Normal
HAAS
Alarm History
Clear

Date	Time	Alarm description

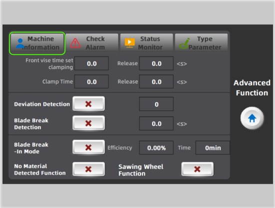
2024 /08 /27 09 : 57 : 30

Alarm Table

CODE	SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
1010.00	No Material Detection Alarm	Material detection switch is broken or has lost signal.	Troubleshoot material detection switch. Refer to **Switch Troubleshooting** section below.
		Vises are improperly clamped.	Check that vises are clamped tightly on material. Inspect vises for chips or excess material.
1000.00 **5201.00**	Oil Pump Stops Automatically.	Machine has not been started for an extended period. Oil pump has automatically shut down.	Start the oil pump.
6000.00	Safety Relay/Emergency Button Engage, Machine Stop.	ESTOP button has been pressed.	Pull up the ESTOP to release.
		ESTOP button is broken or has lost signal.	Confirm that ESTOP is functioning correctly. Refer to **Switch Troubleshooting** section below.
6010.01	Drive Wheel Cover Open.	Drive wheel guard detection switch is mispositioned.	Check the detection distance of the switch. Reposition if necessary.
		Drive wheel guard detection switch has lost signal.	Check function of switch. Refer to **Switch Troubleshooting** section below.
6010.02	Idler Wheel Cover Open.	Idler wheel guard detection switch is mispositioned.	Check the detection distance of the switch. Reposition if necessary.
		Idler wheel guard detection switch has lost signal.	Check function of switch. Refer to **Switch Troubleshooting** section below.
5100.00	Material Width Over Limit.	Incorrect input of material diameter.	Check whether the material diameter is inputted correctly.
5100.01	Material Height Over Limit.	Incorrect input of material height.	Check whether the material height is inputted correctly.
5200.00	No Material Confirmed, Recheck.	Material data is not set.	Set the material data and confirm.
5201.00	Hydraulic Motor Not Started. Start Hydraulic Motor.	Oil pump is not started.	Start the oil pump and proceed with operation.
5202.00	Not Manual Mode, Please Choose Manual Mode.	Incorrect mode.	Toggle switch on control panel to manual.
5203.00	Not Automatic Mode. Please Choose Automatic Mode.	Incorrect mode.	Toggle switch on control panel to automatic.
5102.00	No Size or Quantity. Check Settings.	Size and quantity settings have not been set.	Press material selection button. Confirm that material type, dimensions, cutting length, and number of cuts is set correctly.
5103.00	Parameters Incorrect. Check Settings.	Parameters are outside the system's range.	Verify that paramters are set to reasonable values.

5000.00	Abnormal Upper Limit Detection.	Quick lowering lever is stuck.	Clear any obstructions around quick lowering lever.
		Upper limit switch is mispositioned.	Check the detection distance of the switch. Reposition if necessary.
		Upper limit switch has lost signal.	Check function of switch. Refer to **Switch Troubleshooting** section below.
5002.01 **5002.04**	Limitation of Origin Unusual.	Origin limit switch is mispositioned.	Check the detection distance of the switch. Reposition if necessary.
		Origin limit switch has lost signal.	Check function of switch. Refer to **Switch Troubleshooting** section below.
5104.00 **5105.00**	Buffer Speed Too Slow or Buffer Speed Too Fast	Buffer speed is not set correctly for the shuttle vise.	Refer to the section titled **HAB - Buffer Valve Adjustment** found in the **Solenoid Identification** Chapter for the corrective action.
6200.00	Saw Wheel Inverter Alarm.	Incorrect parameters set on VFD.	Check parameters on VFD. Refer to section **6.5 HAB - Variable Frequency Drive Wiring** to confirm parameters are set correctly. Cycle power and reset the alarm.
6101.00	Coolant Pump Thermal Overload Alarm.	Thermal overload relay malfunction.	Troubleshoot thermal overload relay.
		Pump motor malfunction.	Troubleshoot coolant pump motor.
		Water pump malfunction.	Troubleshoot water pump.
		Tripped breaker.	Refer to the **Tripped Breakers** section below for the corrective action.
6102.00	Hydraulic Pump Thermal Overload Alarm	Thermal overload relay malfunction.	Troubleshoot thermal overload relay.
		Pump motor malfunction.	Troubleshoot oil pump motor.
		Oil circuit malfunction.	Troubleshoot oil circuit.
		Tripped breaker.	Refer to the **Tripped Breakers** section below for the corrective action.
6305.00	Broken Blade Detection Alarm	Driven wheel is stuck or jammed.	Check that driven wheel can rotate normally. Clear any jams or blockages as necessary.
		Switch is mispositioned.	Check detection distance of switch. Reposition if necessary.
		The broken blade detection switch is not reading properly.	Check if switch is functioning correctly. Refer to **Switch Troubleshooting** section below.
5205.00	System is not Initialized.	Bandsaw needs to be homed before beginning operation.	Return to main page and press button **'Saw Frame Return'** to initialize.
---	Intelligent communication anomalies	Software version mismatch	Contaact Haas Service for a resolution. Take photos of the following and submit with the request: • Silver sticker on the PLC • Machine Information screen • machine spec. plate with serial number

Machine Information


Machine Information
Check Alarm
Status Monitor
Type Parameter
Advanced Function

Front vise time set clamping 0.0 Release 0.0 <s>

Clamp Time 0.0 Release 0.0 <s>

Deviation Detection ☒ 0

Blade Break Detection ☒ 0.0 <s>

Blade Break -In Mode ☒ Efficiency 0.00% Time 0min

No Material Detected Function ☒ Sawing Wheel Function ☒

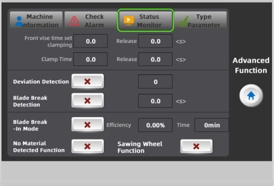
The Machine information page will display the software version of the machine for the HMI and Main Controller

On the version 2 machines the page will also display the PLC version.

Switch Troubleshooting

USING THE STATUS MONITOR

The Status monitor displays the status of each switching signal.


Machine Information
Check Alarm
Status Monitor
Type Parameter
Advanced Function

Front vise time set clamping 0.0 Release 0.0 <S>

Clamp Time 0.0 Release 0.0 <S>

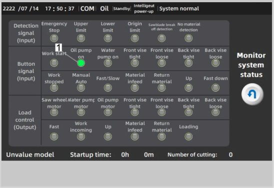
Deviation Detection ☒ 0

Blade Break Detection ☒ 0.0 <S>

Blade Break -In Mode ☒ Efficiency 0.00% Time 0min

No Material Detected Function ☒ Sawing Wheel Function ☒

It is convenient for production operators to monitor the machine status in real time. It is a quick inspection in the event of a failure.


Detection signal (Input)
Emergency Stop
Upper limit
Lower limit
Origin limit
Sawblade break off detection
No material detection
Button signal (Input)
1
Work start
Oil pump on
Water pump on
Front vise tight
Front vise loose
Back vise tight
Back vise loose
Work stopped
Manual Auto
Fast/Slow
Material infeed
Return material
Up
Fast down
Load control (Output)
Saw wheel motor
Water pump motor
Oil pump motor
Front vise tight
Front vise loose
Back vise tight
Back vise loose
Fast
Work incoming
Up
Material infeed
Return material
Loading
Monitor system status

2222 /07 /14 17 : 50 : 37 COM Oil Standby Intelligeut power-up System normal

Unvalue model Startup time: 0h 0m Number of cutting: 0

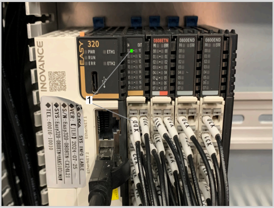
Detection signal (input): The working status indication of the machine tool limit switch and detection switch, and it lights up when the switch is operated (green) (1).

Button signal (input): The machine tool button input status indication, illuminated when the switch is operated (green) (1).

Load control (output): The working status of the motor and solenoid valve is indicated, and it lights up when it is in operation (green) (1).

The status monitor can be used to verify that relevant signal is being transmitted to the HMI, or that a signal is being sent to a relevant component.

USING THE PLC


INOVANCE
EASY 320
1

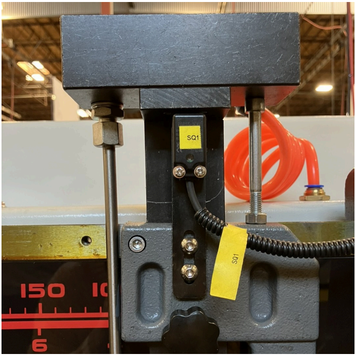
The PLC will show LED lights if the switching signal is activated.

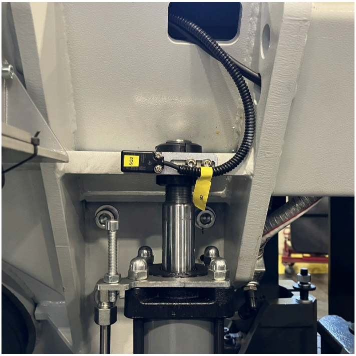
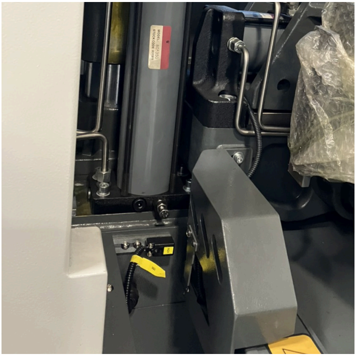
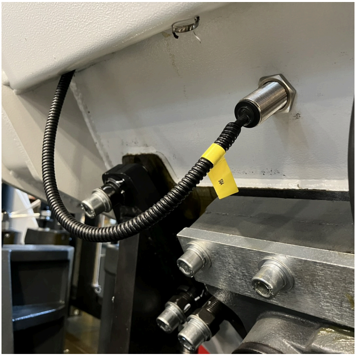
This can be used to indicate that power and signal are being transmitted to the computer from the sensor.

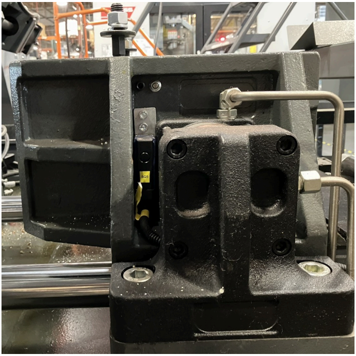
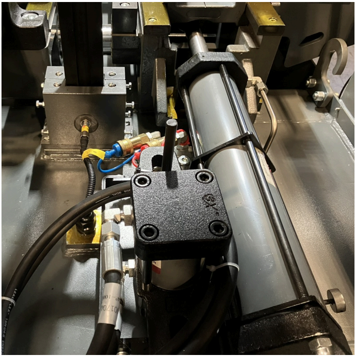
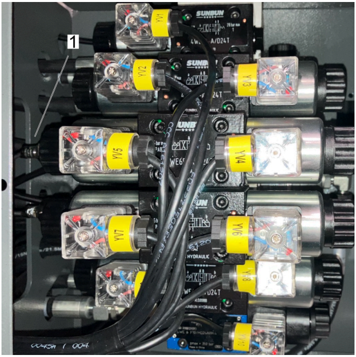
Note: If a switch is manually toggled the LED light should also toggle.

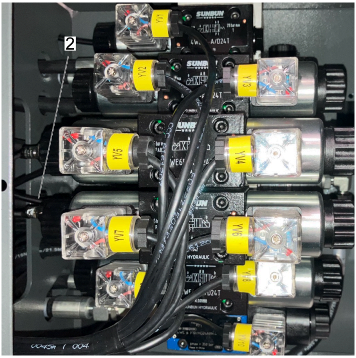
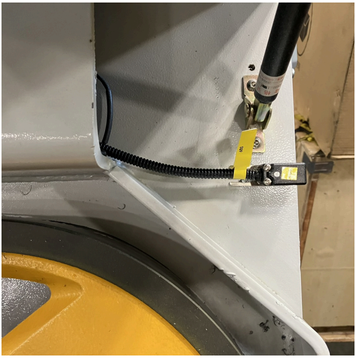
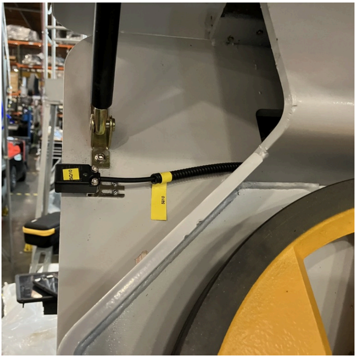
Switch Table

Use the table below to find the location and function of sensors/switches on the HAB-3232.

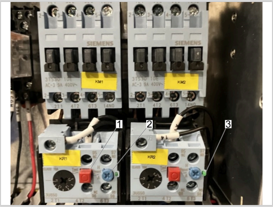
#	SWITCH NAME	LOCATION	IMAGE
SQ1	Initiate Feed Switch	Front of Machine	

SQ2	Lower Limit Switch	Back of Main Pillar (top)	
SQ3	Shuttle Vise Home Switch	Back of Main Pillar (bottom)	
SQ4	Saw Blade Break Off Detection	Back of Drive Wheel Cover	

SQ5	No Material Detection	Rear Vise Limit Switch	
SQ6	Saw Bow Safe Zone Switch	Bottom of Floating End of Vise	
SQ7	Fully Clamped Sensor - YV5 Solenoid	Back of YV5 solenoid [1]	 1 YV1 YV2 YV3 YV5 YV4 YV7 YV6 YV8 SUNBUN

SQ8	Fully Clamped Sensor - YV7 Solenoid	Back of YV7 Solenoid [2]	 2 YV1 YV2 YV3 YV4 YV5 YV6 YV7 YV8 SUNBUN HYDRAULIC
SQ9	Drive Wheel Door Cover Open	Under Drive Wheel Cover	
SQ10	Idle Wheel Door Cover Open	Under Idle Wheel Cover	

Tripped Breakers


SIEMENS
SIEMENS
KM1
KM2
1
2
3
KR1
KR2

1

Corrective action: Check that breakers are set to the on position.

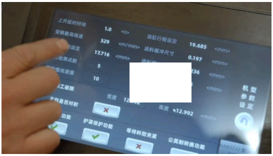
Note: The red indicator will test the breaker[1]. The blue indicator will reset the breaker[2]. If one of the breakers are tripped the green indicator will pop out as shown [3].

Note: The left breaker is for the coolant. The right is for the hydraulic pump.

Close cabinet and turn the circuit breaker is turned to the ON position.

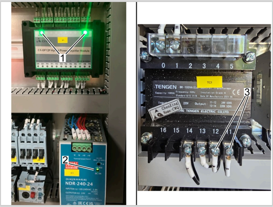
Control Display should light up.

How - To Change Language Settings



Corrective action: Refer to the following video for the proper steps on changing the language of the Haas Automatic Band Saw.

DC Power Supply & Transformer Troubleshooting

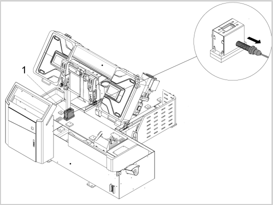

1
2
NDR-240-24
TENGEN
3
0 1 2 3 4 5
16 15 14 13 12 11

Inspect the hybrid amplifier, which operates on both 24V AC and 24V DC. When functioning properly, it will display a green indicator light for each power source—one for AC and one for DC **[1].**

Inspect the DC power supply. When functioning properly a green light on the DC power supply indicates that it is supplying DC voltage **[2].**

Check the transformer's output voltage by measuring terminals 11 and 12 with a multimeter; when functioning properly, they should read 24V AC **[3].**

Proximity Height Sensor Troubleshooting



Inspect the functionality of the HAB3232 proximity height sensor.

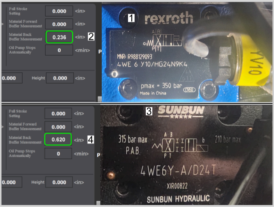
Jog the machine to the top of its travel until the sensor located beneath the head frame **[1]** is triggered.

If the proximity height sensor fails to trigger, loosen the lock nut securing the sensor to allow for adjustment. With the head frame still raised to the top of its travel, carefully thread the sensor inward until it activates. Once triggered, tighten the lock nut to secure the sensor in place and prevent it from shifting.

Important: Thread the sensor in only until it activates. Avoid threading it too close to the saw frame pillar, as this could cause damage to the sensor when raising and lowering the frame.

In the case of failure, replace the proximity height sensor with PN: **93-5589** (V1 saws only)

Solenoid YV10 - Feed Buffer Parameter



Full Stroke Setting 0.000 <in>
Material Forward Buffer Measurement 0.000 <in>
Material Back Buffer Measurement 0.236 <in>
Oil Pump Stops Automatically 0 <min>
0.000 Height 0.000 <in>

Full Stroke Setting 0.000 <in>
Material Forward Buffer Measurement 0.000 <in>
Material Back Buffer Measurement 0.620 <in>
Oil Pump Stops Automatically 0 <min>
0.000 Height 0.000 <in>

1
2
3
4
rexroth
MNR: R988129093
4WE 6 Y10/HG24N9K4
pmax = 350 bar
Made in China
YV10
SUNBUN
315 bar max
P.A.B
210 bar max
4WE6Y-A/D24T
XIR00822
SUNBUN HYDRAULIC

If your machine is experiencing inconsistent/varied cut lengths then the Material Back Buffer Measurement may need to be adjusted.

The Material Back Buffer Measurement will slow the motion of the rear vise by toggling solenoid YV10. this ensures ensure that the next clamping position in accurate.

If the machine has the brand (1) in the photo then the Material Back Buffer Measurement parameter will need to be **0.236in (5.994mm)** as indicated (2).

If the machine has the brand (3) in the photo then the Material Back Buffer Measurement parameter will need to be **0.620in (15.748mm)** as indicated (4).

Note: Only the relevant parameter is shown to have a value for clarity.

6.2 HAB - IDENTIFICATION GUIDE

HAB 3232 - Identification Guide

Revision B - 10/2025

Introduction

The following troubleshooting chapter offers a guide to identifying the three different versions of the HAB3232.

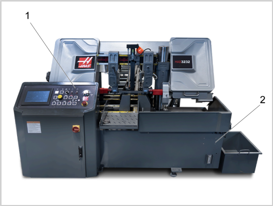
Version 1 bandsaws are contained within the serial following numbers.

90520000 - 90520094

Version 2 bandsaws are contained within the following serial numbers.

90520095 - CURRENT

HAB3232 - Version 1



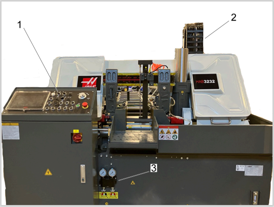
The following image is to help quickly identify the first version of the HAB3232.

The first version of the HAB3232 can be identified by the following listed below:

[1] The electrical control.

[2] The version 1 saw is not equipped with any safety guardings.

HAB3232 - Version 2



1
2
3
HAB3232

The following image is to help quickly identify the second version of the HAB3232.

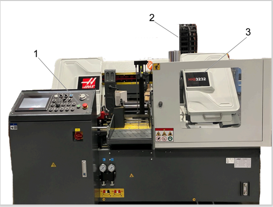
The second version of the HAB3232 can be identified by the following:

[1] The electrical control.

[2] The second version will have the cables routed vertical cable carrier as shown.

[3] The front and rear vise gauges that monitor the vise pressure are mounted on the front of the machine.

HAB3232 - CE


1
2
3
HAAS
HAB3232

The following image is to help quickly identify the CE version of the HAB3232.

[1] The electrical control.

[2] The second version will have the cables routed through a vertical cable carrier as shown.

[3] The CE Bandsaw comes equipped with a sheet metal enclosure that wraps around the machine for protection.

6.3 HAB VERSION 1 - WIRING SCHEMATIC

HAB 3232 - Wiring Diagram

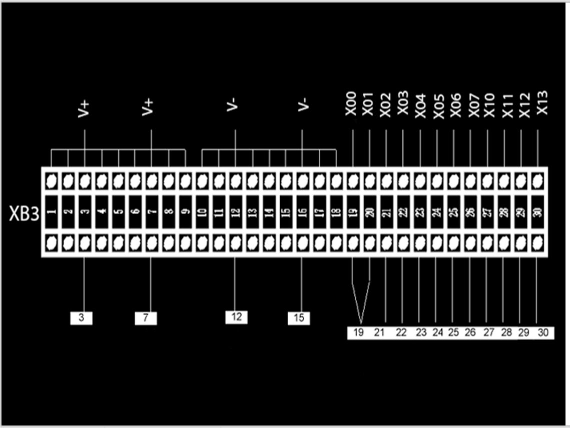
Under Construction

Introduction

HAAS
HAB3232

This section offers an overview of the wiring schematic for the main control of the version 1 Haas Automatic Bandsaw 3232.

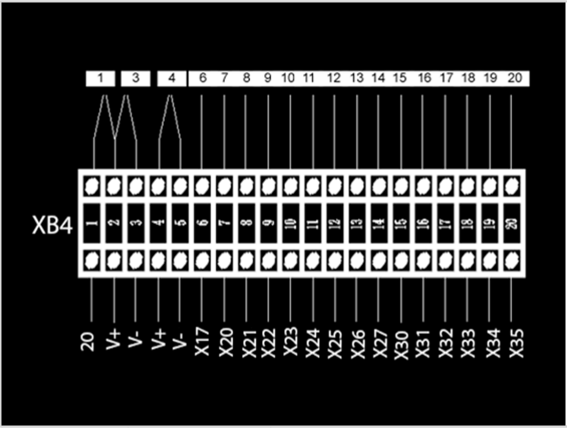
HAB - Main Control Wiring Schematic


V+
V+
V-
V-
X00
X01
X02
X03
X04
X05
X06
X07
X10
X11
X12
X13
XB3
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
3
7
12
15
19 21 22 23 24 25 26 27 28 29 30

GROUP	NUMBER	FUNCTION
	3	Sensor Power Positive
	7	Sensor Power Positive
	12	Sensor Power Negative
	16	Sensor Power Negative

XB3	19	Free Magnetic Signal
	21	Upper Limit Switch
	22	Lower Limit Switch
	23	Feeding Origin Switch
	24	Broken Belt Detection
	25	No Material Detection Switch
	26	Anti-Collision Detection Switch
	27	Main Clamp Pressure Difference Detection
	28	Main Clamp Pressure Difference Detection
	29	Driving Wheel Guard Inspection
	30	Passive Wheel Guard Inspection

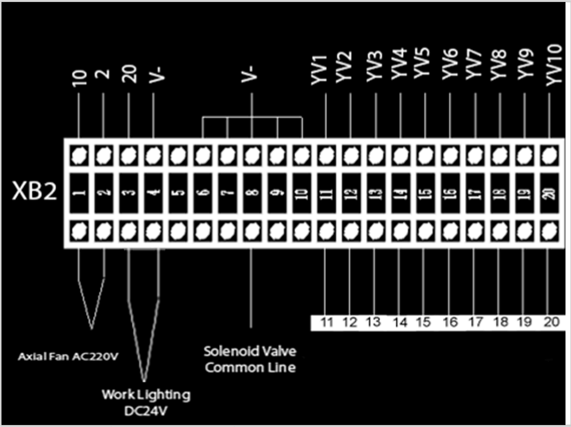
HAB - Main Control Wiring Schematic Continued


1
3
4
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
XB4
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
20
V+
V-
V+
V-
X17
X20
X21
X22
X23
X24
X25
X26
X27
X30
X31
X32
X33
X34
X35

GROUP	NUMBER	FUNCTION
XB4	1	Light Switch Knob
	3	Power Indicator HMI DC 24V
	4	Power Indicator HMI DC 24V
	6	Emergency Stop
	7	Oil Pump Start
	8	Work Start
	9	Work Stop
	10	Saw Lower
	11	Saw Raise
	12	Main Clamping
	13	Main Clamp Release
	14	Clamping Button
	15	Clamp Release
	16	Feed Button

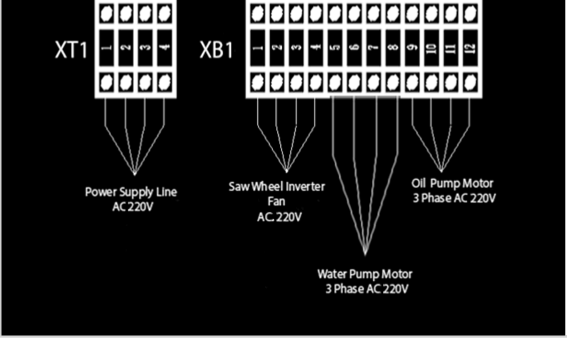
	17	Return Button
	18	Slow & Fast Material Return
	19	Manual / Auto Knob
	20	Water Pump Mode Knob

HAB - Main Control Wiring Schematic Continued


10
2
20
V-
V-
YV1
YV2
YV3
YV4
YV5
YV6
YV7
YV8
YV9
YV10
XB2
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
11 12 13 14 15 16 17 18 19 20
Axial Fan AC220V
Solenoid Valve Common Line
Work Lighting DC24V

GROUP	NUMBER	FUNCTION
XB2	11	Quick Solenoid Valve
	12	Pulling Solenoid Valve
	13	Raising Solenoid Valve
	14	Main Clamping Solenoid Valve
	15	Main Clamping Release Solenoid Valve
	16	Material Clamping Solenoid Valve
	17	Material Clamp Release Solenoid Valve
	18	Feeding Solenoid Valve
	19	Material Return Solenoid Valve
	20	Buffer Solenoid Valve

R
S
T
PE
U2
V2
W2
PE
U2
V2
W2
PE
U3
V3
W3
PE

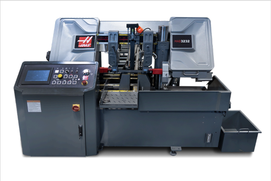

XT1
1
2
3
4
XB1
1
2
3
4
5
6
7
8
9
10
11
12
Power Supply Line
AC 220V
Saw Wheel Inverter
Fan
AC 220V
Oil Pump Motor
3 Phase AC 220V
Water Pump Motor
3 Phase AC 220V

6.4 HAB CE - WIRING SCHEMATIC

HAB 3232 - CE - Wiring Diagram

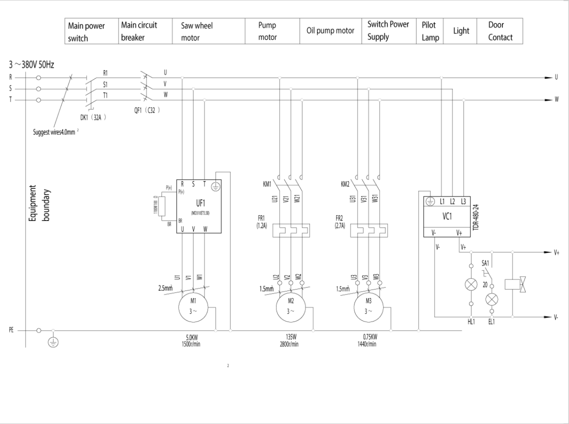
Revision A - 9/2025

Introduction

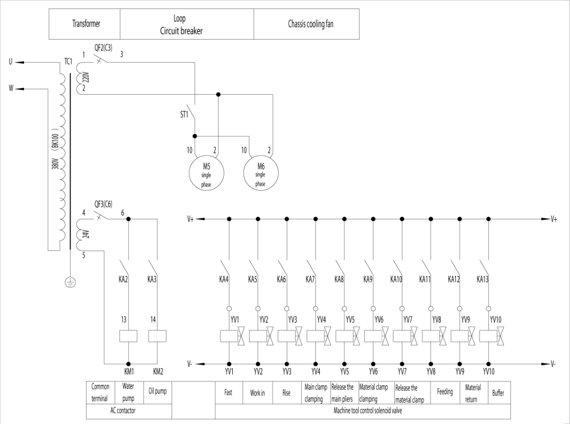


This section offers an overview of the wiring schematic for the CE Version of the Haas Automatic Bandsaw 3232.

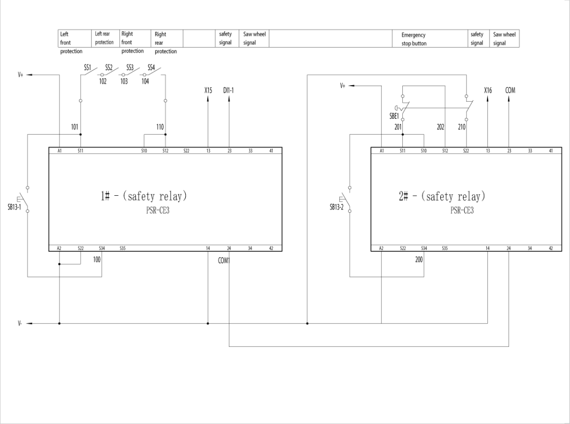
HAB - CE - Main Control Wiring Schematic


Main power switch
Main circuit breaker
Saw wheel motor
Pump motor
Oil pump motor
Switch Power Supply
Pilot Lamp
Light
Door Contact
3～380V 50Hz
R
S
T
R1
S1
T1
U
V
W
DK1（32A）
QF1（C32）
Suggest wires4.0mm²
Equipment boundary
UF1
R S T
P(+)
BR
U V W
KM1
U21 V21 W21
FR1 (1.2A)
KM2
U31 V31 W31
FR2 (2.7A)
VC1
L1 L2 L3
V- V+
TDR-480-24
U1 V1 W1
2.5mm²
M1
3～
5.0KW
1500r/min
U2 V2 W2
1.5mm²
M2
3～
135W
2800r/min
U3 V3 W3
1.5mm²
M3
3～
0.75KW
1440r/min
SA1
20
HL1
EL1
V+
V-
PE
U
W
2

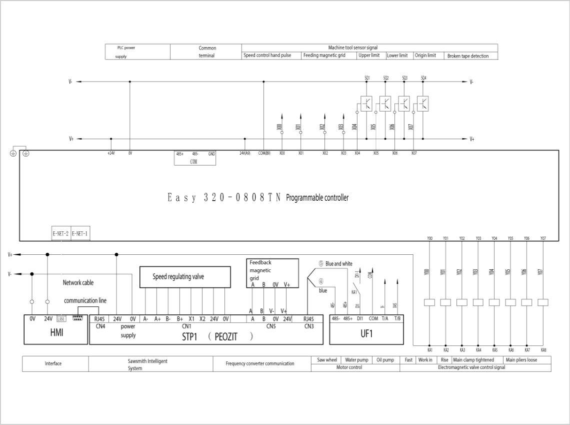
HAB - CE - Main Control Wiring Schematic Cont.


Transformer
Loop Circuit breaker
Chassis cooling fan
U
W
TC1
QF2(C3)
1
3
220V
2
380V (BK100)
ST1
10
2
M5 single phase
10
2
M6 single phase
QF3(C6)
4
6
24V
5
V+
V+
KA2
KA3
KA4
KA5
KA6
KA7
KA8
KA9
KA10
KA11
KA12
KA13
13
14
YV1
YV2
YV3
YV4
YV5
YV6
YV7
YV8
YV9
YV10
V-
V-
KM1
KM2
Common terminal
Water pump
Oil pump
AC contactor
Fast
Work in
Rise
Main clamp clamping
Release the main pliers
Material clamp clamping
Release the material clamp
Feeding
Material return
Buffer
Machine tool control solenoid valve

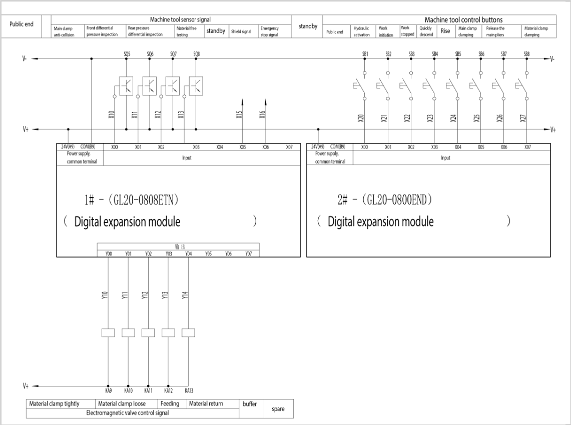
HAB - CE - Safety Relays Wiring Schematic


Left front protection
Left rear protection
Right front protection
Right rear protection
safety signal
Saw wheel signal
Emergency stop button
safety signal
Saw wheel signal
V+
SS1
SS2
SS3
SS4
102
103
104
X15
DI1-1
101
110
A1
S11
S10
S12
S22
13
23
33
41
1# - (safety relay)
PSR-CE3
SB13-1
A2
S22
S34
S35
14
24
34
42
100
COM1
V+
SBE1
201
202
210
X16
COM
2# - (safety relay)
PSR-CE3
SB13-2
200
V-

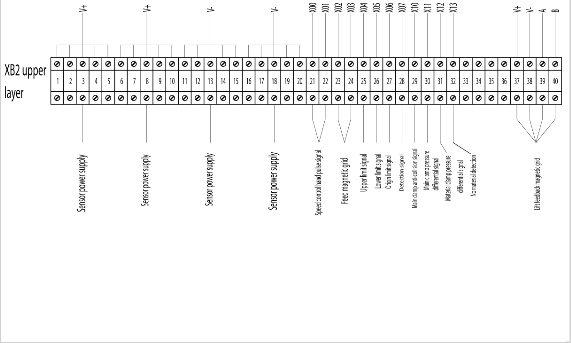
HAB - CE - PLC Wiring Schematic


PLC power supply
Common terminal
Machine tool sensor signal
Speed control hand pulse
Feeding magnetic grid
Upper limit
Lower limit
Origin limit
Broken tape detection
SQ1
SQ2
SQ3
SQ4
V-
V+
+24V
0V
485+
485-
GND
COM
24V(A9)
COM(B9)
X00
X01
X02
X03
X04
X05
X06
X07
E a s y 3 2 0 - 0 8 0 8 T N Programmable controller
E-NET-2
E-NET-1
Y00
Y01
Y02
Y03
Y04
Y05
Y06
Y07
Network cable
communication line
Speed regulating valve
Feedback magnetic grid
A B 0V V+
⑤ Blue and white
④ blue
DI1
COM
KA1
0V
24V
LAN1
HMI
RJ45
CN4
24V
0V
power supply
A-
A+
B-
B+
X1
X2
24V
0V
CN1
STP1 （PEOZIT ）
A
B
V-
V+
0V
24V
CN5
RJ45
CN3
485-
485+
DI1
COM
T/A
T/B
UF1
KA1
KA2
KA3
KA4
KA5
KA6
KA7
KA8
Interface
Sawsmith Intelligent System
Frequency converter communication
Saw wheel
Water pump
Oil pump
Motor control
Fast
Work in
Rise
Main clamp tightened
Main pliers loose
Electromagnetic valve control signal

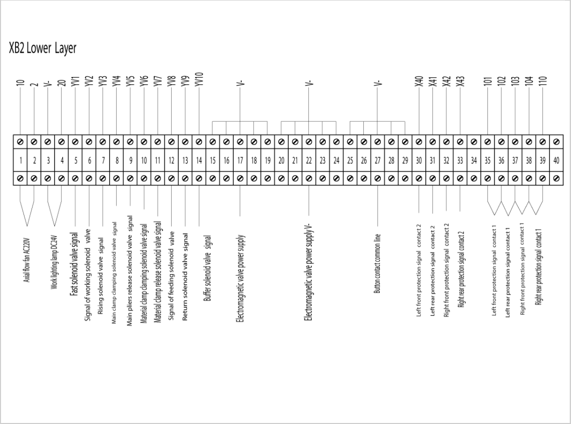
HAB - CE - PLC Wiring Schematic Cont.


Public end
Machine tool sensor signal
Main clamp anti-collision
Front differential pressure inspection
Rear pressure differential inspection
Material free testing
standby
Shield signal
Emergency stop signal
standby
Machine tool control buttons
Public end
Hydraulic activation
Work initiation
Work stopped
Quickly descend
Rise
Main clamp clamping
Release the main pliers
Material clamp clamping
V-
V+
SQ5
SQ6
SQ7
SQ8
X10
X11
X12
X13
X15
X16
SB1
SB2
SB3
SB4
SB5
SB6
SB7
SB8
X20
X21
X22
X23
X24
X25
X26
X27
24V(A9)
COM(B9)
X00
X01
X02
X03
X04
X05
X06
X07
Power supply, common terminal
Input
1# －（GL20-0808ETN）
（ Digital expansion module ）
2# －（GL20-0800END）
（ Digital expansion module ）
输 出
Y00
Y01
Y02
Y03
Y04
Y05
Y06
Y07
Y10
Y11
Y12
Y13
Y14
KA9
KA10
KA11
KA12
KA13
Material clamp tightly
Material clamp loose
Feeding
Material return
Electromagnetic valve control signal
buffer
spare

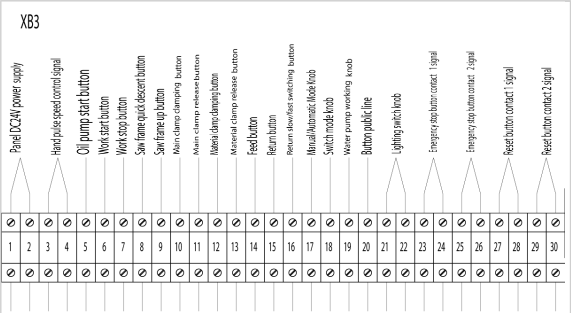
HAB - CE - Push In Terminal Block Wiring Schematic


XB2 upper
layer
V+
V+
V-
V-
X00
X01
X02
X03
X04
X05
X06
X07
X10
X11
X12
X13
V+
V-
A
B
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
Sensor power supply
Sensor power supply
Sensor power supply
Sensor power supply
Speed control hand pulse signal
Feed magnetic grid
Upper limit signal
Lower limit signal
Origin limit signal
Detection signal
Main clamp anti-collision signal
Main clamp pressure differential signal
Material clamp pressure differential signal
No material detection
Lift feedback magnetic grid

HAB - CE - Push In Terminal Block Wiring Schematic Cont.

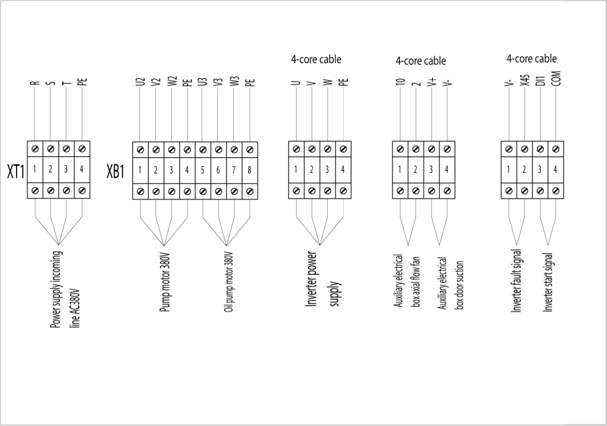

XB2 Lower Layer
10
2
V-
20
YV1
YV2
YV3
YV4
YV5
YV6
YV7
YV8
YV9
YV10
V-
V-
V-
X40
X41
X42
X43
101
102
103
104
110
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
Axial flow fan AC220V
Work lighting lamp DC24V
Fast solenoid valve signal
Signal of working solenoid valve
Rising solenoid valve signal
Main clamp clamping solenoid valve signal
Main pliers release solenoid valve signal
Material clamp clamping solenoid valve signal
Material clamp release solenoid valve signal
Signal of feeding solenoid valve
Return solenoid valve signal
Buffer solenoid valve signal
Electromagnetic valve power supply
Electromagnetic valve power supply V-
Button contact common line
Left front protection signal contact 2
Left rear protection signal contact 2
Right front protection signal contact 2
Right rear protection signal contact 2
Left front protection signal contact 1
Left rear protection signal contact 1
Right front protection signal contact 1
Right rear protection signal contact 1

HAB - CE - Push In Terminal Block Wiring Schematic Cont.


XB3
Panel DC24V power supply
Hand pulse speed control signal
Oil pump start button
Work start button
Work stop button
Saw frame quick descent button
Saw frame up button
Main clamp clamping button
Main clamp release button
Material clamp clamping button
Material clamp release button
Feed button
Return button
Return slow/fast switching button
Manual/Automatic Mode Knob
Switch mode knob
Water pump working knob
Button public line
Lighting switch knob
Emergency stop button contact 1 signal
Emergency stop button contact 2 signal
Reset button contact 1 signal
Reset button contact 2 signal
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

V+
V-
X00
X01
X20
X21
X22
X23
X24
X25
X26
X27
X30
X31
X32
X33
X34
X35
X36
V-
20
V+
201
202
210
V-
100
101
200
201

HAB - CE - Push In Terminal Block Wiring Schematic Cont.


XT1
R
S
T
PE
1
2
3
4
Power supply incoming line AC380V
XB1
U2
V2
W2
PE
U3
V3
W3
PE
1
2
3
4
5
6
7
8
Pump motor 380V
Oil pump motor 380V
4-core cable
U
V
W
PE
1
2
3
4
Inverter power supply
4-core cable
10
2
V+
V-
1
2
3
4
Auxiliary electrical box axial flow fan
Auxiliary electrical box door suction
4-core cable
V-
X45
DI1
COM
1
2
3
4
Inverter fault signal
Inverter start signal

6.5 HAB - VARIABLE FREQUENCY DRIVE WIRING

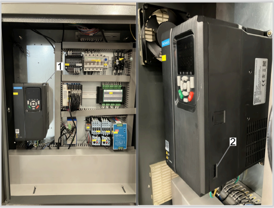
HAB 3232 - Variable Frequency Drive Wiring

Revision B - 6/2025

Introduction

The following chapter provides a guide to ensure the correct wiring of the Haas Automatic Band Saw Variable Frequency Drive (VFD).

VFD - Main Terminal Wiring Inspection

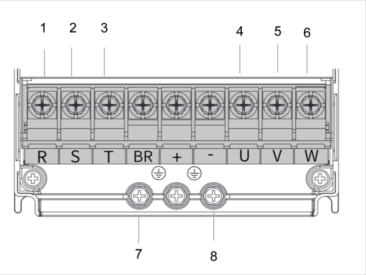


Power **OFF** the machine and set the main circuit breaker to the **OFF** position.

Open the control cabinet of the machine and locate the VFD [1].

Remove the front cover by using the two push tabs found on both sides of the VFD [2].

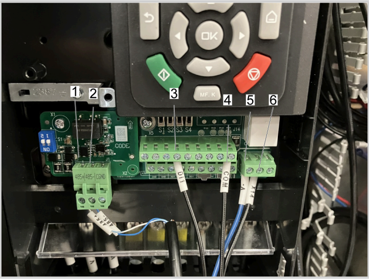
Refer to the chart below to connect each properly labeled wire to its corresponding terminal.


1
2
3
4
5
6
R
S
T
BR
+
-
U
V
W
7
8

NUMBER / TERMINAL	CABLE LABEL
1 / (R)	U
2 / (S)	V
3 / (T)	W
4 / (U)	U1
5 / (V)	V1
6 / (W)	W1
7	PE
8	PE

VFD - PCB Wiring Inspection

Refer to the chart below to connect each properly labeled wire to its corresponding terminal.

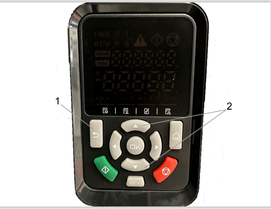

1
2
3
4
5
6
OK
CODE
S1 S2 S3 S4
485+|485-|CGND
DI1
COM

NUMBER / TERMINAL	CABLE LABEL	TERMINAL NAME
1	485+	458+
2	485-	485-
3	D11	D11
4	COM	COM
5	V-	T/A
6	X14	T/B

VFD - Parameter Inspection and Tuning

Ensure that the VFD is Programmed correctly.

NOTE: VFD parameters must be verified when replacing the VFD.



1

Open the electrical control panel and turn on the machine.

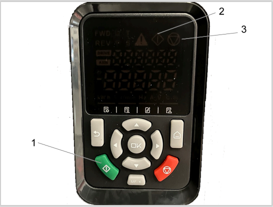
To access the operations panel on the VFD , click the return arrow key **[1]**.

⚠**Note:** Program the VFD to align with the setting values in the chart below

Navigate through different parameters using the up and down arrow keys **[2]**.

Press **OK** to modify the parameter.

FUNCTION CODE	FUNCTION DESCRIPTION	FACTORY DEFAULT	SETTING VALUE
F0-01	The first motor control mode	2	0
F0-02	Command Source Selection	0	0
F0-09	Moving direction	0	0
F0-10	Maximum frequency	50	200
F0-12	Upper limiting frequency	50.0	200
F0-17	Acceleration time 1	20.0	3.0
F0-18	Deceleration time1	20.0	3.0
F1-00	Motor type	2	2
F1-01	Rated power of motor	7.5	5.0
F1-02	Rated voltage of motor	220	220
F1-03	Rated current of motor	19.0	17.9
F1-04	Rated frequency of motor	50	133.3
F1-05	Rated motor speed	1500	2000
Fd-00	Baud rate	5005	5007
Fd-01	Data format	0	3



2

After entering the values from the chart above, go to parameter F1-37 and set it to 11.

Press the green confirm key **[1].**

Note: Motor tuning will begin. During the process, the light will turn green [2]. Once tuning is complete, the light will turn red [3].

After tuning is complete, navigate to parameter F0-02 and set its value to 1. Additionally, set the value of parameter F0-03 from 0 to 9.

6.3 HAB - HYDRAULIC SYSTEM

HAB 3232 - Hydraulic System

Revision E - 08/2026

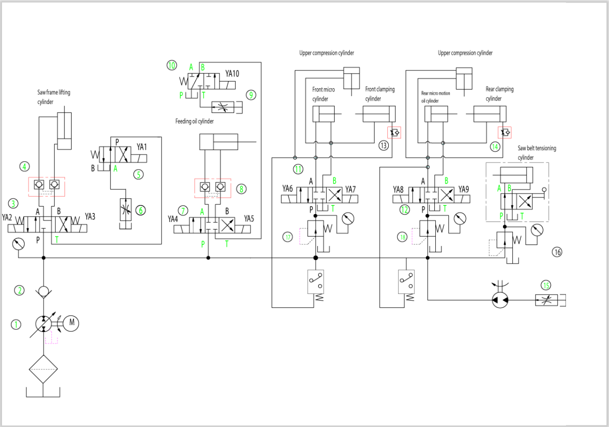
Introduction

The following troubleshooting chapter offers a guide to identifying and adjusting the hydraulic sytem of the HAB3232.

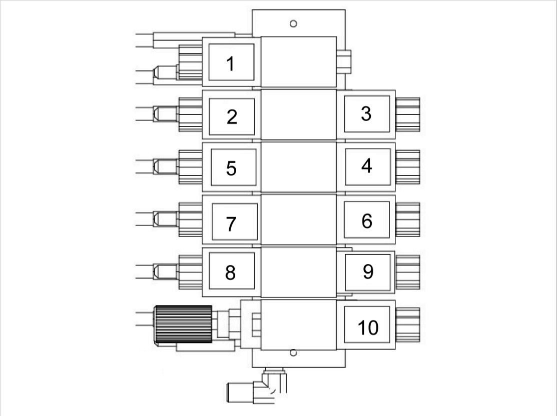
Table of Contents:

- Hydraulic Diagram
- Solenoid Identifications
- Pressure Regulation
- Buffer Valve Adjustment

HAB - Hydraulic Diagram


Upper compression cylinder
Upper compression cylinder
Front micro cylinder
Front clamping cylinder
Rear micro motion oil cylinder
Rear clamping cylinder
Saw belt tensioning cylinder
Saw frame lifting cylinder
Feeding oil cylinder
YA1
YA2
YA3
YA4
YA5
YA6
YA7
YA8
YA9
YA10
①
②
③
④
⑤
⑥
⑦
⑧
⑨
⑩
⑪
⑫
⑬
⑭
⑮
⑯
⑰
⑱
A
B
P
T
W
M

HAB - Solenoid Identification Chart

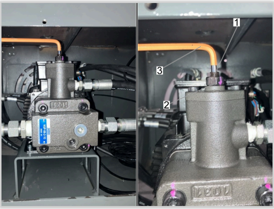

1
2
3
5
4
7
6
8
9
10

NUMBER	SOLENOID	FUNCTION	DESCRIPTION
1	YA1	Rapid	Saw frame rapid movement.
2	YA2	Saw Frame Lower	Actuates saw frame to lower.
3	YA3	Saw Raise	Raises the saw blade vertically.
4	YA4	Front Vise Open	Actuates front vise piston to open.
5	YA5	Front Vise Clamp	Actuates front vise piston to close.
6	YA6	Rear Vise Clamp	Actuates both the long travel pistons and the short travel pistons to close the rear vise.
7	YA7	Rear Vise Open	Actuates both the long travel piston and the short travel piston to open the rear vise.
8	YA8	Material Infeed	Actuates forward movement of the rear vise.
9	YA9	Return Material	Actuates rearward movement of the rear vise.

10	YA10	Feeding Buffer	Slows the rapid of the rear vise when approaching the correct length of the next part and when approaching the blade.

HAB - Pressure Regulation

SYSTEM PRESSURE



To adjust the pressure, tighten/loosen the pressure relief set screw **[1]** until the operating pressure is between **3.8 - 4.1 MPA.**

Break loose the locknut **[2]** so that the set screw can be adjusted.

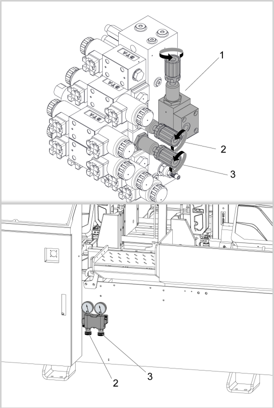
To increase the operating pressure, tighten the set screw **[1]** with an 5mm Allen key **[3]**. To decrease the operating pressure, loosen the set screw.

Note: Make small adjustments and give a few seconds in between each adjustment so that the system has time to build to the new operating pressure.

Once you have the operating pressure in the correct range, turn off the pump and allow it to sit for 30-60 seconds. Then power the pump back up to verify that the pressure builds to the same MPA it was set at.

Tighten the locknut **[2]** to hold the set screw in position.

VISE PRESSURE AND BLADE TENSION



SOLENOID	FUNCTION	SPEC (MPA)
1	Blade Tension	V1 - 3.8-4.1 V2 - 3.6 - 3.9
2	Front Vise	2.0-3.9
3	Rear Vise	2.0-3.9

The front and rear vise pressure adjustment knobs on the Version 1 HAB-3232 bandsaw are located on the solenoid manifold. This manifold located in a cabinet on the left side of the machine.

The front and rear vise pressure adjustment knobs on the Version 2 HAB-3232 bandsaw are located at the front of the machine. This manifold is noted in the second photo in this section.

The blade tension gage is located on the saw bow above the blade tensioning piston.

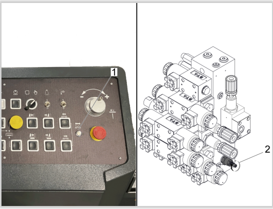
Note: Refer to the BLADE INSTALLATION CHAPTER of the HAB Manual for more information on regards to adjusting the blade tension .

HAB - Buffer Valve Adjustment

The purpose of the buffer valve is to slow the rate of travel of the rear vise.

The rear vise must be slowed in automatic mode when the rear vise is moving backward, approaching the set length of the part, and when the vise is moving forward, positioning that set length under the blade. This is what allows the hydraulic system to retain its part length accuracy.

Note: The buffer valve should only be adjusted in the scenario in which the alarm **Buffer Speed Too Slow or Buffer Speed Too Fast** is present.

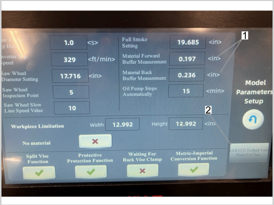


On the control press the Material Speed Switch **[1]** to activate the buffering switch on the bandsaw**.**

Use a 14mm crescent wrench to loosen the lock nut found on the adjustment valve.

When jogging the rear vise forward or backward, use a 4mm Allen wrench to turn the knob **[2]** clockwise to increase the positioning speed, or counterclockwise to decrease it.

Set the buffer speed to a point at which the alarm clears on the control.



1.0 <s>

nverter Top peed 329 <ft/min>

Saw Wheel Diameter Setting 17.716 <in>

Saw Wheel Inspection Point 5

Saw Wheel Slow Line Speed Value 10

Full Stroke Setting 19.685 <in>

Material Forward Buffer Measurement 0.197 <in>

Material Back Buffer Measurement 0.236 <in>

Oil Pump Stops Automatically 15 <min>

Workpiece Limitation Width 12.992 Height 12.992 <in>

No material ✗

Split Vise Function ✓

Protective Protection Function ✓

Waiting For Back Vise Clamp ✗

Metric-Imperial Conversion Function ✓

Model Parameters Setup
1
2

Ensure that the Buffer Material Forward and Material Back distances are set to the following values **[1].**

Note: Press the reset button **[2]** to clear the current values and restore the original settings.